

SEARCHING 3

"One finds limits by pushing them."

~ Herbert Simon



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Good
Evening ☺

Today's content

- Aⁿ magical no.
- Painter's partition

Doubt class on sunday → 11 AM

TOH ⇒ Recording available in post
lecture content of Recursion 3

Q Find A^{th} magical no.

A number is magical if it is divisible by B or C.

Eg 1 :- $B=2$, $C=3$ $A=8$ $\rightarrow$ 8th magical no.

1 2 3 4 5 6 7 8 9 10 11 12
1 2 3 4 5 6 7 8
 $\rightarrow$ Ans = 12

Eg 2 :- $B=2$, $C=7$, $A=3$ $\rightarrow$ 3rd magical no.

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16
1 2 3
 $\rightarrow$ Ans = 6

Eg 3 :- $B=17$ $C=3$ $A=5^{\text{th}}$ magical no.

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16
1 2 3 4
 $\rightarrow$ Ans = 15

Brute force $\rightarrow$ Consider all the numbers from 1 until count $\leq A$

increase count if a no. is divisible by B or a number is divisible by C

$$\text{Time complexity} = [1 \text{ to } A * \min(B, C)]$$

$$\text{No of iterations} = A * \min(B, C)$$

$$\text{Max possible Range} = A * \min(B, C)$$

$$\text{Min possible Range} = 1$$

$$\text{Range of ans} = \{1 \quad A * \min(B, C)\}$$

Search space

$$\text{Target} = A * \text{magical number}$$

Let's discuss a smaller problem

Q. Given P, Q, r . Find the count of magical no. from 1 to r

$$P=3 \quad Q=5 \quad r=35$$

Ans = Calculate the count of magical no. [1 35]

3 5 6 9 10 12 15 18 20 21 24 25 27 30 33 35

$$\text{Ans} = 16$$

$$\text{Multiples of 3 in range } [1, 35] = \frac{35}{3} = 11$$

$$\text{Multiples of 5 in range } [1, 35] = \frac{35}{5} = 7$$

$$\text{Multiples of } 3 \times 5 \text{ in range } [1, 35] = \frac{35}{3 \times 5} = 2$$

$$\text{Final ans} = 11 + 7 - 2 = 16$$

02. Count of magical no. in range $[1, 100]$

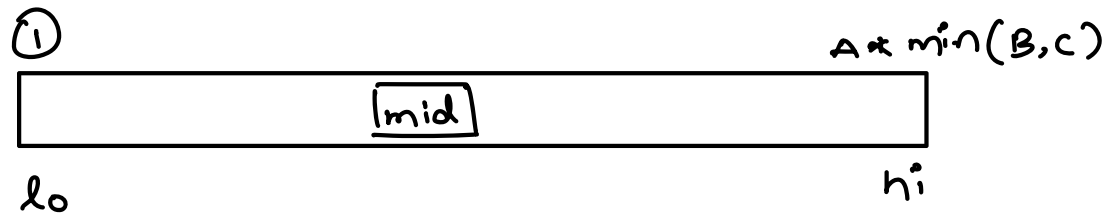
$$p=9, q=12$$

$$\text{Ans} = \frac{100}{9} + \frac{100}{12} - \frac{100}{108}$$

$$= \frac{100}{9} + \frac{100}{12} - \frac{100}{\text{LCM}(9,12)} = 11 + 8 - 2 = 17$$

$$\text{Search space} = \{1 \quad A * \min(B, C)\}$$

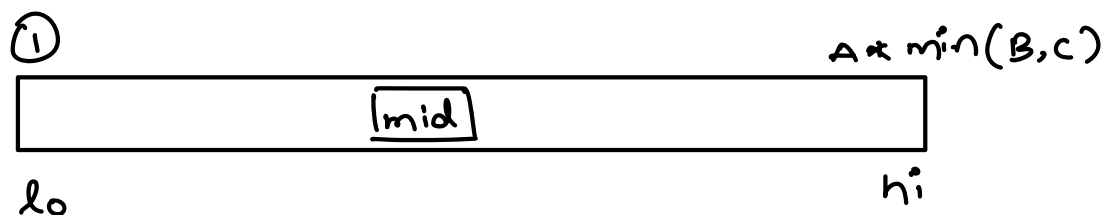
Target = A^{th} magical number



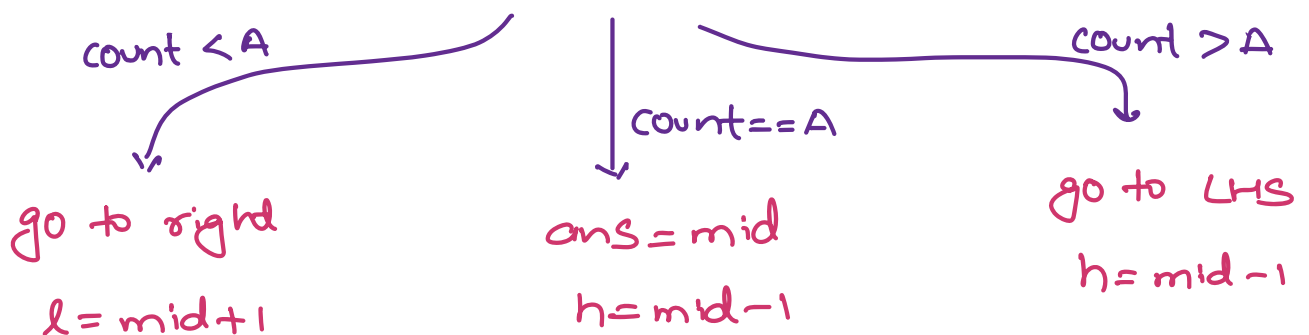
$B=5 \quad C=7 \quad A=3 \rightarrow 3^{\text{rd}}$ magical no.

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	0	0	0	1	1	2	2	2	3	3	3	3	4	5

count of magical no.



count the magical no. $\leq mid$



$$B=5 \quad C=7 \quad A=3$$

$$\text{search space} = \{1 \quad 15\}$$

$$\text{target} = 3^{\text{rd}} \text{ magical no.}$$

$\frac{lo}{1}$	$\frac{hi}{15}$	$\frac{mid}{8}$
----------------	-----------------	-----------------

$$\text{count the magical no. } [1, mid]$$

$$\frac{8}{5} + \frac{8}{7} - \frac{8}{35} = 2$$

$$lo = mid + 1$$

9	15	12
---	----	----

$$\frac{12}{5} + \frac{12}{7} - \frac{12}{35} = 3$$

$$\text{ans} = 12$$

$$h = mid - 1$$

9	11	10
---	----	----

$$\frac{10}{5} + \frac{10}{7} - \frac{10}{35} = 3$$

$$\text{ans} = 10$$

$$h = mid - 1$$

9	9	9
---	---	---

$$\frac{9}{5} + \frac{9}{7} - \frac{9}{35} = 2$$

$$lo = mid + 1$$

10 9

Exhausted search space

$$\text{Ans} = 10$$

$l = 1 \quad h = A * \min(B, c)$

while ($l \leq h$)

mid = $(l + h) / 2$;

count = countmagical(B, c, mid)

if ($\text{count} == A$)

ans = mid ;

h = mid - 1

}

else if ($\text{count} < A$)

l = mid + 1

}

else

h = mid - 1 ;

}

return ans

TC : $O(\log A * \min(B, c) * (\log \max(B, c)))$

SC : $O(1)$

int count_magical (int B, int c, int mid)

int ans = 0

$$\text{ans} = \frac{\text{mid}}{B} + \frac{\text{mid}}{c}$$

int gcd = GCD (B, c)

int lcm = B * c / gcd

$$\text{ans} = \text{ans} - \frac{\text{mid}}{\text{lcm}}$$

return ans;

3

Complete concept
of GCD on
sunday

int GCD (int a, int b)

if (b == 0) return a

return GCD (b, a % b)

3

TC : $O(\log_2 \max(a, b))$

$$\text{LCM}(a,b) = \frac{a * b}{\text{GCD}(a,b)}$$

10:12 pm → 10:20 pm

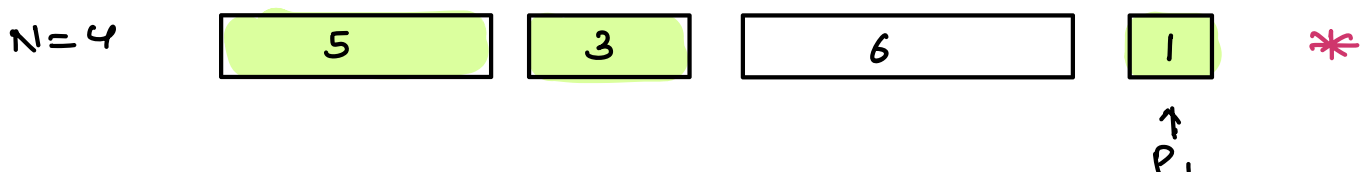
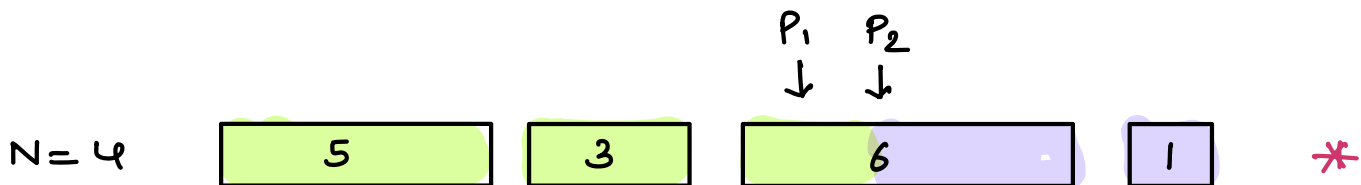
02. Painter's partition problem - I

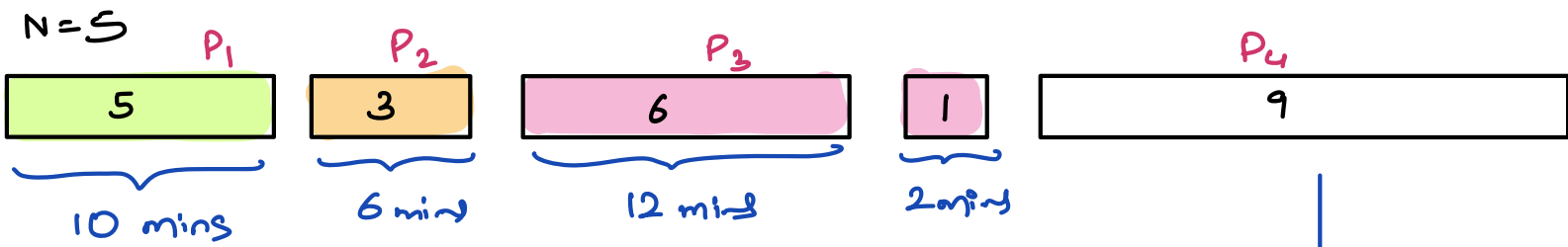
Given N boards with length of each board

- (a) A painter takes T unit of time to paint 1 unit of length
- (b) A board can only be painted by 1 painter
- (c) A painter can only paint boards placed next to each other (i.e. continuous segment)

Q Find minimum no. of painters required to paint all boards in X unit of time. Return -1 if not possible.

N=5





$T = 2$ unit of time to paint 1 unit of length

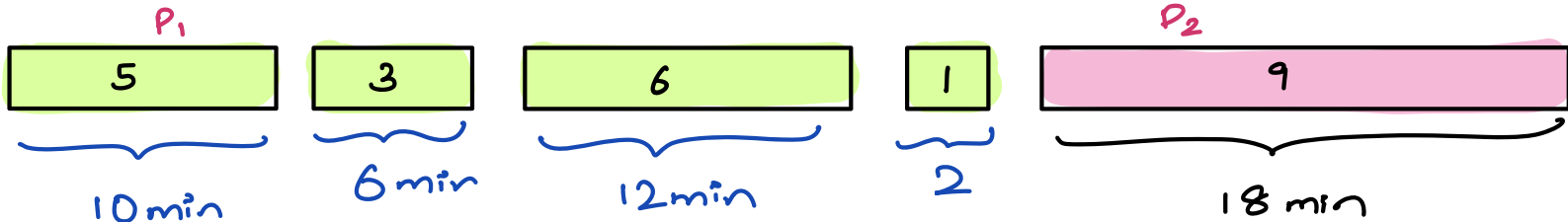
$X = 15$ mins (total unit of time)

Max permissible time = 15 min

But P_4 at least needs 18 min to paint it

Ans = -1

$N=5$ $T=2$ $X=30$ mins

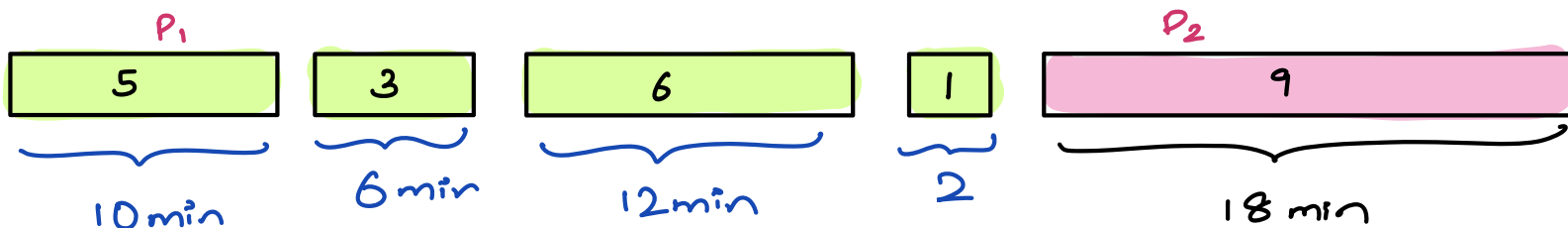


$$P_1 = 10 \text{ min} + 6 \text{ min} + 12 \text{ min} + 2 \text{ min}$$

$$P_2 = 18 \text{ min}$$

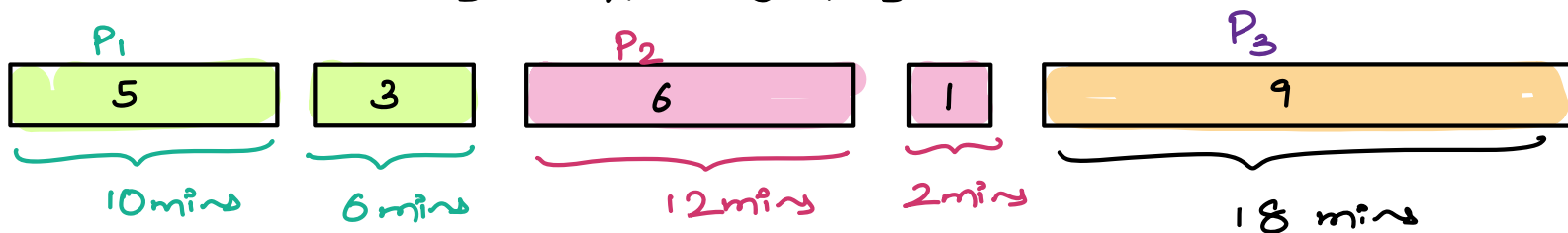
Ans = 2

$N=5$ $T=2$ $X=32 \text{ mins}$



Ans = 2

$N=5$ $T=2 \text{ mins}$ $X=20 \text{ mins}$



Ans = 3 painters

```
int minpainter (int A[], int t, int x)
```

```
    painter = 1 , timeleft = x
```

```
    for (i=0; i<n; i++)
```

```
        if (A[i] * t > x) return -1
```

```
        if (A[i] * t <= timeleft)
```

```
            | timeleft = timeleft - A[i] * t;
```

```
        } else {
```

```
            painter ++;
```

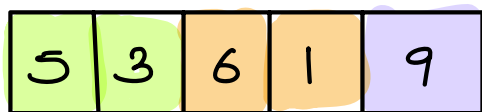
```
            timeleft = x - A[i] * t
```

```
        }
```

```
    }
```

return painter

3



$$x = 20$$

$$k = 2$$

$$P = 1$$

~~2~~

3

$$\underline{\text{time left}} = 20 - 18 = 2$$

$$9 + 2 \leq 6 \text{ min}$$

$$\textcircled{18} \leq 6$$

Painter Partition II

Q Find minimum time to paint all boards if P painters are available.

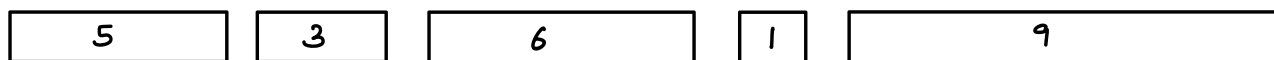
$N=5$



$P = \text{painters}$

min time = ?

$N=5$



$P=1$

$T = 2 \text{ mins for}$
1 unit of length

$$\begin{aligned} \text{Ans} &= 5 * 2 + 3 * 2 + 6 * 2 + 1 * 2 + 9 * 2 \\ &= (5 + 3 + 6 + 1 + 9) * 2 \end{aligned}$$

$$= 24 * 2 = 48 \text{ mins}$$

			Total time
P = 2 T = 2	P ₁ = 1	→ 5 * 2 = 10 mins	38 mins
	P ₂ = next 4 boards	→ (3 + 6 + 1 + 9) * 2 = 38	
	P ₁ = 2	→ (5 + 3) * 2 = 16 mins	32 mins
	P ₂ = 3	→ (6 + 1 + 9) * 2 = 32 mins	
	P ₁ = 3	→ (5 + 3 + 6) * 2 = 28 mins	28 mins
P ₂ = 2	→ (1 + 9) * 2 = 20 mins		
	P ₁ = 4	→ (5 + 3 + 6 + 1) * 2 = 30 mins	20 mins
	P ₂ = 1	→ (9) * 2 = 18 mins	

Max possible time → Only 1 painter

$$\rightarrow \sum A[i] * t$$

Min possible time = 1 or maxele * t

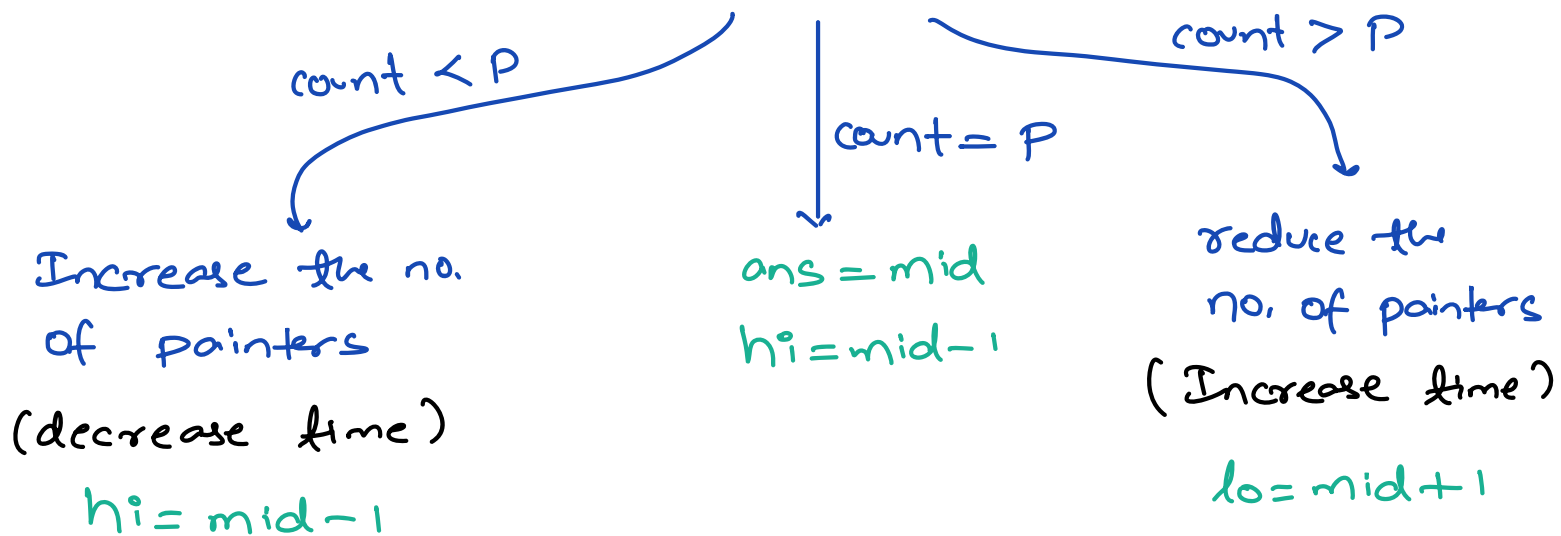
$$\text{Search space} = [\text{maxele} * t \quad \sum A[i] * t]$$

Target = Min time with P painters

Binary search on time



count of painters required to
paint all the boards in
time = mid



1 painter = max time

2 painter = ↓

3 painter = ↓

No. of painters $\propto \frac{1}{\text{time}}$

$$N=5$$

$$P=2$$

$$T=2$$

5	3	6	1	9
---	---	---	---	---

P_1

low hi mid count no. of painters = mid time

18 48 33 2 ans = 33 mins
h = mid - 1

18 32 25 $3 > P$ l = mid + 1

26 32 29 $2 == P$ ans = 29
h = mid - 1

26 28 27 $3 > P$ l = mid + 1

28 28 28 $2 == P$ ans = 28
h = mid - 1

28 27

Search space is finished

Ans = 28

Code $\rightarrow$ { TODO }

int mintime (A[], t, p)

lo = max A[i] * t

hi = $\sum A[i] * t$

while (lo $\leq$ hi)

mid = (lo + hi) / 2

painter = minpainter(A, t, mid)

if (painter == p)

ans = mid

hi = mid - 1

else if (painter < p)

hi = mid - 1

else {

lo = mid + 1

return ans

$$TC: O(\text{search space}) * n$$

$$TC: O\left(\sum A[i] * \ell - \max A[i] * \ell + 1\right) * n$$

$$SC: O(1)$$

$$\text{mid} = \frac{\text{lo} + \text{hi}}{2}$$

$$= \text{lo} + \frac{(\text{hi} - \text{lo})}{2}$$

$$= \frac{2\text{lo} + \text{hi} - \text{lo}}{2} = \frac{2\text{lo} - \text{lo} + \text{hi}}{2} = \frac{\text{lo} + \text{hi}}{2}$$

$$TC = O(n \log n * \underbrace{\tau_c}_{\text{of factors}})$$

$$SC = O(1)$$

} Look at
Google

Floor of a no.